

Signal-Scope — Model Report

Real vs Synthetic Image Detection | SIH-2026 Internal Hackathon

1. Task

Binary image classification: predict whether an image is **likely AI-generated** or **likely real**, with a confidence score. The interface uses likelihood language rather than a definitive claim.

2. Data & Split

Training source	CIFAKE-style real/synthetic dataset; 100,000 labelled images (50,000 FAKE + 50,000 REAL).
Train / validation	80,000 / 20,000; random split with seed 42.
Local evaluation	20,000 images (10,000 FAKE + 10,000 REAL), kept separate from train/validation.
Input resolution	Source images were 32×32; inference accepts larger images and resizes them to 224×224.

3. Model / Approach

Backbone: EfficientNet-B0 pretrained on ImageNet with transfer learning. The final classifier was replaced by a 2-class linear layer. **Preprocessing:** Resize 224×224 + ToTensor. **Loss:** CrossEntropyLoss. **Optimizer:** Adam, learning rate 1×10⁻⁴. **Batch size:** 64. **Epochs:** 3. **Augmentation:** none in the reported run. **Calibration:** no separate calibration; displayed confidence is the softmax probability. **Threshold:** 0.50.

4. Training Results

Epoch	Train Loss	Train Acc.	Val Loss	Val Acc.
1	0.1301	94.85%	0.0650	97.64%
2	0.0564	97.87%	0.0547	98.04%
3	0.0341	98.75%	0.0594	97.87%

5. Evaluation Results

Metric	Result
Accuracy	98.07%
Macro Precision	98.07%
Macro Recall	98.07%
Macro F1	98.07%
ROC-AUC	0.9981
Average Precision	0.9981
Decision threshold	0.50
FPR @ 0.50	Not retained in the recorded run
Confusion matrix	Exact class-wise counts not retained in the recorded run
Official unseen-generator AUC	Not available from current local evaluation; organizer-held-out score

6. Baseline & Limitations

Baseline: A separate numerical score for the challenge-provided baseline was not recorded, so no baseline comparison is claimed. **Limitations:** the model was trained on low-resolution 32×32 source images and cannot recover fine detail absent from training data. Larger uploads are accepted but resized to 224×224. No augmentation, explicit calibration, frequency-domain features, Grad-CAM explanation, robustness testing, or generator-attribution module was included in this baseline. Manual testing also observed a false positive on a real image, so outputs should be treated as likelihoods.

Submission note: The problem statement requires overall AUC, unseen-generator AUC, macro-F1, accuracy/FPR at a stated threshold, and a confusion matrix. The official unseen-generator score is organizer-evaluated and is intentionally not fabricated here.